

## Subject: Bill Densmore's "vibe" chat with Claude Sonnet 4.6

Date: Sunday, Feb. 21, 2026 | 9:50 p.m. to approx. 12:30 a.m. 02-22-26

This is a copy-and-paste of my entire interaction with Anthropic's Claude Sonnet 4.6 free model. To distinguish my comments from Claude's I have left mine in the Segoe UI sans-serif font and Claude's extensive comments in the Georgia (serif) font. I have made only two edits: (1) put "BILL:" in front of my comments to further distinguish and (2) added selective underlining of passages I consider significant. I have not edited my questions/instructions or Claude's responses in any way other than a handful of misspellings on my part. By uploading documents gradually, I allowed Claude to suggest its own advice of logical sequence which matches reality remarkably well. You will see as you read this chat that it asks for things and makes suggestions that I "feed" to it and ultimately it decides it has enough to write a funder pitch and tech specification without more input. Claude initially says a tech spec doc is required, that would take months, then goes on to write a 21-pager after I upload all the documents we have at hand. It astounds me that this wasn't written by a person. The replies came back so fast, and the written documents, as to make that impossible.

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BILL: I want to stand up a shared-user network for trust, identity, privacy and digital information commerce -- a free market for digital information. It involves four parties: A cloud authentication service, a user home base, a plurality of publishers and a logging, settlement and billing service in the cloud. Users need to be able to have one account at a home base and be recognized by other publisher sites on the network, using some form of Single Sign On (SSO) through the authentication service. The protocols and policies need to be governed by a 501(c)3 nonprofit. How do you suggest I get started?

9:52 PM

This is a well-scoped vision — essentially a federated identity and information commerce network with nonprofit governance. It has parallels to efforts like OpenID, OASIS, the Trust Framework initiatives, and more recently initiatives like MyData.org and Solid (Tim Berners-Lee's project). Here's how I'd think about getting started:

### **Start with governance, not technology**

The nonprofit is your most important first move, not the tech stack. Everything else — protocol design, membership agreements, billing rules — derives legitimacy from the governance structure. A 501(c)3 focused on "educational and charitable purposes" around digital rights and open standards is the right vehicle.

You'll want to:

File for incorporation in a state favorable to nonprofits (Delaware or California are common). Draft bylaws that define membership classes — you'll likely want founding members, publisher members, and possibly consumer advocates. Stand up a technical steering committee and a policy committee as separate bodies from day one, so that protocol decisions and business rule decisions don't collapse into each other. This separation is what makes a standards body trustworthy to all four parties.

### **Map the trust framework before writing a line of code**

A "trust framework" is the legal and policy layer that sits above the protocols. It defines what a certified home base must do, what a publisher must agree to, how the authentication service is audited, and what settlement rules the billing service must follow. Look at the NSTIC (National Strategy for Trusted Identities in Cyberspace) trust framework model and the Open Identity Exchange (OIX) for templates. You need this document before you can onboard any parties, because it's what they're actually signing when they join the network.

### **Choose your identity federation architecture deliberately**

You have real options here, each with different tradeoffs. OpenID Connect (OIDC) on top of OAuth 2.0 is the dominant SSO protocol today and has massive ecosystem support — most publishers' engineering teams already know it. SAML is older and enterprise-heavy, still common in B2B contexts. Solid/WebID is more privacy-preserving and user-centric but has a smaller ecosystem. For a digital information commerce network where user consent and data portability matter, I'd strongly lean toward OIDC as the core protocol with verifiable credentials (W3C VC standard) layered on for richer identity and consent claims. This lets the home base issue credentials the user controls, not just tokens the auth service controls.

### **Think carefully about the logging, settlement and billing service**

This is the part most identity efforts underinvest in and where commercial viability lives. You need to define early: what constitutes a billable event (article read, dataset accessed, API call), how revenue splits are calculated between home base and publisher, how disputes are handled, and what audit trail is required. The settlement service is essentially a clearing house, similar to how payment networks work. Immutable append-only logs (not necessarily blockchain — a well-designed database with cryptographic audit trails works fine) are important here for trust between parties.

### **Recruit founding members strategically**

You need at least one publisher, one potential home base operator, and one technically credible authentication service partner willing to sit at the table before you can credibly design the protocols — because the protocols need to reflect real operational constraints. Look at academic publishing networks, independent media consortia, or library systems as natural early publisher partners since they

already have shared-identity infrastructure thinking and mission alignment with a nonprofit model.

### **Reference architectures worth studying closely**

The Internet2 and InCommon federation (higher ed identity federation) is probably the closest operational analog to what you're describing. Project VRM (Vendor Relationship Management, Doc Searls) maps closely to the commerce and user-agency side. MyData.org's operator model handles the home base concept well. Stripe's approach to marketplace settlement is worth borrowing from for the billing layer even if the technology differs.

The sequencing I'd suggest is: incorporate the nonprofit and draft governance bylaws → develop the trust framework document → recruit 3-5 founding member organizations → convene a working group to draft protocol specifications → build a reference implementation → pilot with founding members → open membership.

The whole process realistically takes 18-36 months to reach a credible pilot. The governance and trust framework work is slower than the technology and more important to get right.

What aspect would you like to go deeper on first — the governance structure, the technical architecture, or the trust framework design?

BILL: We're already incorporated and have 501(c)3 status with the IRS since 2017. It is the Information Trust Exchange Governing Association, or

<https://www.itega.org> The bylaws are here: <https://newshare.com/itega-irs-calif/itega-restated-bylaws-ADOPTED-01-27-2020.pdf>

9:57 PM

Great — let me read what you actually have before saying anything else.

Searched the web

Searched the web

Let me also look at the projects and [vision pages](#) to understand where ITEGA currently stands.

Searched the web

Searched the web



Good — I've now read the [bylaws](#) in full and the vision page carefully. Here's an honest, grounded assessment of where you are and what the most useful next steps look like.

### **What you've already built well**

The governance foundation is solid. The bylaws are lean and functional — director terms, electronic voting procedures, conflict of interest requirements, the separation of fiscal and board functions. The "no statutory members" provision in Article V is a deliberate design choice that keeps decision-making concentrated in the board while you're in early formation, which is appropriate. The ICANN-for-identity framing on the vision page is exactly right as a mental model and a pitch. The technical vision on the site is also appropriately specific — OIDC plus Shibboleth-like federation, cohort-based advertising anonymization, opt-in privacy-by-design. That's not vague hand-waving; those are buildable things.

### **What I'd flag as the current critical gap**

The site's most recent substantive content appears to be from 2021-2022. The "Privacy Beat" newsletter and the 2019 conference suggest there was real momentum, but the question is whether there's an active technical working group and active publisher/operator commitments right now, or whether the organization is still primarily in advocacy and research mode. The difference matters enormously for what the right next step is.

### **The core problem you need to solve to move from vision to network**

A federated identity network has a classic cold-start problem — publishers won't integrate until there are users, and users won't create accounts until publishers are integrated. The way every successful federation has broken this is by finding an anchor institution or consortium that brings both sides at once. InCommon worked because universities are simultaneously identity providers and service providers. What's ITEGA's equivalent forcing function?

The most natural answer, given your journalism focus and the Local Media Consortium relationship visible on the site, is a regional news consortium that agrees to be both the first "home base" operator and the first publisher cohort simultaneously. That lets you run a real pilot with real users inside a contained, mission-aligned community before opening to the broader network.

### **Concrete next steps I'd prioritize**

The first thing I'd do is produce a current-state technical specification document — even a 10-15 page working draft — that translates the vision page into an actual protocol spec: what the authentication service must implement, what the home base operator API looks like, what events the settlement service logs, what the membership agreement obligates publishers to do. This document becomes the artifact around which you recruit technical partners and anchor institutions. Without it, every conversation starts over from scratch.

Second, I'd revisit Article V of the bylaws. The "no statutory members" provision made sense in early formation, but to build a genuine multi-stakeholder network you'll eventually need a formal membership structure — publisher members, operator members, possibly user advocate seats — with defined rights and obligations. The bylaws allow amendment with a 4/5 board vote, so this is doable when the time is right.

Third, the settlement and billing layer deserves a standalone design document. This is the piece that most directly enables the "commerce" part of your information commerce vision, and it's also the piece most likely to generate sustainable revenue for ITEGA itself through transaction fees. It needs a clear answer to: what is the minimum viable clearing function, and what existing payment or micropayment infrastructure could ITEGA ride on top of rather than build from scratch?

### **What I'd want to know from you**

Is ITEGA currently operating with active staff, an active board, and relationships with potential anchor publishers — and looking for strategic direction on how to sequence the technical buildout? Or is the more immediate need to re-activate and re-resource the organization itself? The answer shapes everything about what useful help looks like.

BILL: A company I founded: Clickshare Service Corp. (<https://www.clickshare.com>) obtained a patent on this approach but the patent, granted in about 2008, has expired for term so its ideas are in the public domain now. We stood up a reference implementation in 1996 but were never able to get marketing traction because at that time the news industry was convinced it would get rich on advertising and didn't need to charge for content. Unfortunately Google and Facebook cornered the market on digital advertising. Here is a link to the patent. Recognize that the tech is ancient but the discussion of purpose and relationships is relevant:

<https://drive.google.com/file/d/1VWgtivJiuYFew97InVV5khh1PvlasPK4/view?usp=sharing>

10:06 PM

Let me read the patent carefully.

Fetches: clickshare-four-party-patent-01-29-2008.pdf

Fetches: clickshare-four-party-patent-01-29-2008.pdf



The Google Drive link requires authentication, so I can't fetch it directly. Let me try to find the patent through the USPTO or Google Patents instead.

Searched the web

Searched the web



Good — I was able to pull the full patent abstract and key technical language from Google Patents. Let me now read the full patent text directly.

Fetches: Managing transactions on a network: four or more parties

Fetches: Managing transactions on a network: four or more parties



Now I have a thorough understanding of the full system. This is a remarkably rich document — essentially a complete system design written in 1997-98 that anticipated nearly everything the modern federated identity and micropayments

ecosystem is still struggling to achieve. Let me give you a substantive, grounded response.

Having read the patent in full, a few things stand out clearly.

### **What you built was genuinely ahead of its time — and the core architecture is still sound**

The TVS (Token Validation Service) model described in this patent is structurally almost identical to what OpenID Connect does today, except that you designed it in 1997 with session tokens, distributed CALS nodes, a home-provider concept, anonymous user IDs that only the home provider can resolve, and ACH-based settlement. The "markup ratio" field — allowing different service providers to charge different retail prices for the same content unit — is a feature that no current federated identity system has implemented. That's a genuinely novel commerce primitive that still doesn't exist in any deployed network at scale. The key architectural insight that holds up perfectly: the system operates by identifier numbers, not by names, and the service itself cannot correlate an ID with a person — only the user's chosen home base can make that association. Google Patents That is privacy-by-design in its purest form, and it's what GDPR, CCPA and every privacy regulation since has been groping toward as a requirement.

### **What's different now versus 1997 that works in your favor**

Three things have changed that make this far more buildable today than it was then:

First, the standards layer now exists. OIDC, OAuth 2.0, JWT tokens, and W3C Verifiable Credentials give you a protocol foundation that maps almost directly onto the TVS architecture. Your token model, home-provider concept, and session management are all expressible in standard OIDC flows. You don't need to build a C-language API or custom wire protocol — you can compose this from existing libraries.

Second, the settlement infrastructure is vastly better. Your patent referenced ACH via Bank of Boston as the settlement backbone. Today you have Stripe Connect, modern ACH APIs, and stablecoin rails as options for the settlement layer, all with developer-friendly interfaces. The micropayment problem (5 cents and up, as the patent notes) is still not perfectly solved, but it's far more tractable than in 1997.

Third, the regulatory and market environment has shifted decisively your way. GDPR, CCPA, the collapse of third-party cookies, and growing publisher desperation for revenue alternatives mean the problems your system was designed to solve are now widely recognized as urgent. The news industry didn't want this in 1997 because advertising felt infinite. It is now in genuine crisis.

### **The honest gap between where ITEGA is and where it needs to be**

Reading the patent alongside the ITEGA website, the intellectual foundation is fully developed but the operational layer is missing. What you need to go from "well-articulated vision with expired patent" to "live network" is essentially this stack:

- A current-language protocol specification that translates TVS into OIDC + W3C VC terms. The patent's architecture maps cleanly, but a modern implementer needs a document written in the vocabulary of 2025, not 1997. This is probably a 20-30 page technical spec, achievable with 2-3 months of focused work by someone who knows both the original system and modern identity standards.

- A reference implementation in a modern stack. The original was an NCSA httpd patch in C. The modern equivalent would be a Docker-deployable OIDC provider with a home-base operator SDK, a publisher plugin (ideally a WordPress plugin given the news industry's infrastructure), and a settlement service API. This is probably a 6-month engineering effort for a small team.

- An anchor consortium commitment. As I mentioned before, the cold-start problem is your real enemy. The strongest move here, given your journalism focus and the Clickshare relationships with Boston Globe, Omaha World-Herald, Sun-Times and Crain Communications, is to go back to those or comparable publishers and ask for a formal commitment to a pilot — not a "we're interested" but a signed letter of intent to implement if a working reference implementation exists.

### **What I'd actually do next, concretely**

The most leveraged single action available to you right now is probably writing a "ITEGA Technical Architecture v1.0" document — essentially a translation of the TVS patent into modern protocol terms, with explicit mappings to OIDC, VC, and current settlement APIs. This document would serve simultaneously as a recruiting tool for technical contributors, a specification for a reference implementation, and a public demonstration of institutional seriousness. It would also be the document you'd submit to a foundation like Mozilla, Knight, or MacArthur for a grant to fund the reference implementation.

Do you want help drafting that architecture document? Given what's in this patent, much of the structure already exists — it's largely a translation exercise with some modernization decisions to make explicitly.

BILL: I will respond momentarily to your suggestion of developing a tech spec that uses modern tools. But first I don't want to leave hanging your question earlier about whether ITEGA is staffed or not. As you suggest, it needs to be re-activated and re-resourced, although the corporation remains active and in good legal standing. My thought is that we need to rally a small consortium of

foundations and academic institutions, probably hosted by the Donald W. Reynolds Journalism Institute (<https://www.rjionline.org>). The Missouri Press Association would participate with the idea of putting in place a proof-of-concept service involving a small number of independent Missouri newspapers. We have a U.S.-registered service mark: "Newshare" which might be appropriate. Your thoughts on this before we get back to the tech spec idea?

10:18 PM

Let me look at RJI and the Missouri Press Association quickly before responding.

Fetchd: Home

Fetchd: Home



Good — that's enough context. Let me give you a substantive response.

### **On the RJI hosting idea — this is a strong instinct**

RJI is an excellent candidate for several reasons. It sits inside the Missouri School of Journalism at Mizzou, which gives it institutional credibility, academic infrastructure, and student/fellow labor that can contribute to a proof-of-concept. Its fellowship program — which explicitly funds "practical and innovative resources for community-centered news" — is a direct fit for what you're describing. Notably, RJI already has an active partnership with the Local Media Consortium on the News Media Help Desk, which means there's an existing organizational relationship with the LMC, a body that has historically been supportive of ITEGA. That's not a coincidence you should ignore — it's a warm path in.

The hosting relationship you'd want isn't necessarily RJI as a legal host of ITEGA (which remains its own 501(c)3), but rather RJI as the institutional home for the proof-of-concept project — providing facilities, academic legitimacy, fellowship-funded talent, and a neutral convening identity. This matters because independent Missouri newspapers will be more willing to participate in something anchored at Mizzou than in something anchored at a small nonprofit they've never heard of, even a well-intentioned one.

### **The Missouri Press Association angle is smart but needs a realistic scope conversation**

The MPA as a co-convenor gives you access to the publisher community without having to cold-recruit. But you need to be honest with yourself about what "a small number of independent Missouri newspapers" means operationally. I'd suggest thinking in terms of three to five papers with active, digitally-engaged audiences — ideally a mix of a small daily and two or three weeklies or digital-only outlets. The goal of the proof-of-concept isn't to demonstrate scale; it's to demonstrate that the four-party architecture actually works end-to-end: a user creates one account at a home base, authenticates, visits a second participating publisher, gets recognized without re-registering, and a transaction record is

logged and settled. If you can demonstrate that clean loop with real users and real publishers, you have something fundable and replicable.

### **On the Newshare service mark — yes, use it**

"Newshare" is better than "ITEGA" for a consumer-facing and publisher-facing brand in this context for a few reasons. ITEGA is a governance and standards body — it's the right name for the nonprofit that sets the rules, but it's not a compelling name for a network that publishers and readers will join. "Newshare" has the right connotations: sharing news, sharing access, sharing value. It signals the journalism focus immediately. Keeping ITEGA as the governing body and Newshare as the network/service brand is exactly the right two-layer structure — analogous to how ICANN governs the DNS but consumers just know their domain registrar.

### **The foundation/academic consortium you need**

For re-activation funding, the most likely sources in rough priority order are:

- the Knight Foundation (has funded journalism technology infrastructure repeatedly and has existing awareness of this problem space),
- the Reynolds endowment itself through RJI (if RJI is the host, internal project funding is possible),
- the MacArthur Foundation (funds information ecosystem work),
- and potentially the Lenfest Institute (focused specifically on local news sustainability).

A consortium of two or three of these alongside RJI as institutional host would give you the \$300K-\$500K needed for an 18-month proof-of-concept with a small technical team and the Missouri pilot.

The ask to foundations should be framed not as "fund a startup" but as "fund a public-interest technical standard and reference implementation that the journalism industry can adopt without vendor lock-in." That framing is well-matched to foundation priorities and distinguishes it clearly from commercial competitors like Piano or Pico.

### **The sequence I'd recommend**

Get a meeting with RJI's director to explore the hosting conversation before approaching foundations. If RJI is in, they become a co-applicant on grants, which significantly strengthens any foundation application. The MPA conversation can happen in parallel. Then, with those two institutional relationships in hand, develop a concise 4-6 page project brief — not a full grant proposal yet, just enough to have a serious conversation with Knight and Lenfest.

That brief is also the moment where having a clear technical spec summary (the document I suggested earlier) becomes very useful, because funders will ask "what exactly are you building."

Shall we work on that project brief, or return to the technical architecture document first? They're related enough that doing a sketch of both together might be the most efficient path.

BILL: Well I'm human and it is now 10:30 p.m. so I'm going to knock off for the night soon. I need to know how to bookmark the exchange we have had so far because it is spot on and super helpful. I am mightily impressed at the work your human engineers and your own "intelligence" is able to do. So before we wrap it for the night, let me link you to a tech specs document which exists. Also, RJI's executive director, is a supporter of this project and they have funded it over the years. I was an academic year RJI Fellow at Mizzou in 2008-2009, See: <https://rjionline.org/person/bill-densmore/>

10:27 PM

Let me take a quick look at your RJI profile, and then I'll address the bookmarking question and the tech spec link.

Fetches: Bill Densmore

Fetches: Bill Densmore



Good — now I have the full picture of who you are and the depth of the RJI relationship. This is much better than a cold introduction — you're an alumnus fellow with a documented track record there, and the work is a direct continuation of your RJI research. That's a genuinely strong re-entry point.

**On bookmarking this conversation:** The easiest way is to use your browser's bookmark function right now on this page (Ctrl+D on Windows, Cmd+D on Mac). If you're using Claude.ai, this conversation should also be saved automatically in your conversation history in the left sidebar, so you can return to it by title. You can also use the share button if you want a link to send to collaborators.

**On the tech spec document** — please do share that link when you're ready. We can pick up there tomorrow or whenever suits you.

A few closing thoughts for tonight: what strikes me most reviewing your RJI profile is that the through-line from 1993 to now is completely coherent. You've been working on the same essential problem — how independent publishers survive and serve communities in a digital economy — for over thirty years, through Clickshare, the fellowship, CircLabs, Taxonometrics, and now ITEGA. That's not a liability, it's your credibility. When you walk back into RJI to have this conversation, you're not pitching a new idea — you're returning to complete work that institution helped start.

Get some rest. This is a genuinely important project and we've made real progress mapping it tonight. Looking forward to seeing the tech spec tomorrow.

BILL: I'm going to stay up a few more minutes to get you some of the tech-spec type documents. It will take about 10 minutes so hang tight Messr. Claud.

10:29 PM

Take your time — I'll be right here. Looking forward to seeing what you have.

BILL: OK Claude I'm going to link you to a bunch of documents with description URLs. I want you to digest them and then create as close to the tech spec necessary to discuss with technical partners and potential funders. Here we go:

10:38 PM

Ready when you are — send the links and descriptions. I'll read everything carefully before writing a word of the spec.

BILL: OK, here's the first one. Tell me if you can read it:

[https://drive.google.com/file/d/1\\_LbRVu9doGIESyaPnoZg2jFHM3d\\_ZeVR/view?usp=drive\\_link](https://drive.google.com/file/d/1_LbRVu9doGIESyaPnoZg2jFHM3d_ZeVR/view?usp=drive_link)

10:39 PM

Failed to fetch:

[https://drive.google.com/file/d/1\\_LbRVu9doGIESyaPnoZg2jFHM3d\\_ZeVR/view?usp=drive\\_link](https://drive.google.com/file/d/1_LbRVu9doGIESyaPnoZg2jFHM3d_ZeVR/view?usp=drive_link)

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Unfortunately I'm hitting a 401 authentication error — the file requires a Google account login to access, so I can't fetch it directly. There are a couple of ways around this:

1. **Change the sharing settings** on each Google Drive file to "Anyone with the link can view" (no sign-in required). That's the most reliable fix.
2. **Export and paste** the text content directly into the chat if the documents aren't too long.
3. **Upload the files directly** to this conversation using the attachment/paperclip button in the chat interface — that will make them immediately accessible to me without any sharing permissions needed.

Option 3 is probably the fastest if you have the files on your computer. Option 1 is best if you want the links themselves to keep working for future collaborators too. Your call — take whichever path is easiest at this hour.

OK, try it now:

[https://drive.google.com/file/d/1\\_LbRVu9doGIESyaPnoZg2jFHM3d\\_ZeV?usp=sharing](https://drive.google.com/file/d/1_LbRVu9doGIESyaPnoZg2jFHM3d_ZeV?usp=sharing)

10:39 PM

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Fetches: itega-technical-design-specs-v6-01-09-26.pdf



Still hitting the Google sign-in wall — it's rendering the login page rather than the document itself, even with the sharing link. This is a limitation on my end; I can't authenticate with Google. The most reliable path forward is to **upload the PDF files directly** into this chat using the paperclip/attachment button. That bypasses Google Drive entirely and I can read the full document immediately. Alternatively, if the document is a Google Doc (rather than an uploaded PDF), you could try **File → Download → Plain Text (.txt)** and paste the content here, or share a direct download link by replacing `/view` with `/export?format=txt` in the URL for Google Docs. Given that it's late, uploading directly is probably the fastest option. The file is called **itega-technical-design-specs-v6-01-09-26.pdf** — if you have that on your computer, just drag it into the chat.

PDF

[https://drive.google.com/file/d/1\\_LbRVu9doGIESyaPnoZg2jZeVR/view?usp=sharing](https://drive.google.com/file/d/1_LbRVu9doGIESyaPnoZg2jZeVR/view?usp=sharing)

10:42 PM

**itega-technical-design-specs-v6-01-09-26.pdf**

Got it — I've read all 23 pages carefully. This is a substantial and well-developed document. Before you send any additional files, let me confirm what I now have in hand and flag what would make the tech spec most complete.

#### **What this document gives me:**

This is genuinely comprehensive at the architectural and requirements level. Sections A through G together cover the four-party structure, the privacy-by-design principles, the attribute exchange schema, the wholesale-retail pricing model, the logging/settlement design, and the functional specs for protocols and network. The RSL standard reference and the Internet2/eduPerson attribute references are particularly useful modern anchors. The document is clearly v6 of something that has been iterated seriously over time.

#### **What it doesn't yet do that a tech spec for 2026 partners and funders needs:**

It doesn't yet map each functional requirement to a specific modern protocol or technology choice. For example, it says "standardized transfer of a unique, non-

repudiable user identifier" — but a technical partner in 2026 needs to see "this is implemented via an OIDC **sub** claim issued by the home-base IdP, carried as a JWT Bearer token." The document describes **what** the system must do with precision; what's needed to complete it is a layer that says **how**, using current standards vocabulary.

**Before I draft the complete spec, please send any remaining documents you have.** Based on your earlier conversation, I know there are references to at least these additional documents that may exist:

- The consumer user-case PDF ([newshare.com/newyork/READING-consumer-user-case.pdf](https://newshare.com/newyork/READING-consumer-user-case.pdf))
- The governance goals document (mentioned as a companion to this one)
- The proof-of-concept prototype elements document
- The user profile attributes spreadsheet

Send whatever you have, and then I'll write the full modernized tech spec document in one pass. Go ahead whenever you're ready.

PDF

BILL: OK Claude please next read the itega-single-sign-on ... network-description:

10:49 PM

Got it — I've read the SSO document fully. This is Richard Lerner's technical contribution and it's the most implementation-specific piece yet. A few things it adds that are critically important:

The **anonymous network user ID scheme** is particularly elegant and important — the home IdSP maintains a mapping from its internal userID to a unique **networkUserId** for each remote server the user visits. So REMOTE-1 and REMOTE-2 each see a different opaque identifier for the same user, making cross-site correlation technically impossible without the home base's cooperation. This is more privacy-protective than standard OIDC's **sub** claim, which is typically the same across a relying party's session. It maps closely to what the W3C calls a **pairwise pseudonymous identifier** — and is actually specified in OIDC's PPID extension. That's a very clean modern hook.

The **NetworkGroupId** bitmask table is also a concrete, implementable artifact — defining subscription tiers (anonymous, registered, print subscriber, digital subscriber, paid, trial, etc.) as combinable bit flags. That's directly usable in a JWT claims schema today.

The **10-step authentication flow** is essentially a standard OIDC Authorization Code flow with a home-site discovery step added — which maps to what the OIDC ecosystem calls **WebFinger** or **identity provider discovery**.

**Please send any remaining documents now** — I have enough to write a very strong spec, but I want everything you have before I start drafting. What else is coming?

BILL: More coming in a few minutes. BTW Rick Lerner is CEO of Clickshare Service Corp. and he is a Carnegie-Mellon computer science Ph.D. Here is his mini-bio:

<https://www.clickshare.com/about/team>

10:54 PM

Good to know — that's a significant asset. A Carnegie Mellon CS PhD who is also the CEO of Clickshare and has been living with this architecture since 1996 is exactly the kind of technical co-founder credibility that both engineering partners and funders will respond to seriously. The fact that Lerner's fingerprints are already on the SSO document — and that it's technically precise enough to map directly to modern OIDC specs — confirms he's been thinking about this in implementable terms, not just conceptual ones.

Fetches: Key team members

Fetches: Key team members



The bio text didn't render fully — just his photo and contact info came through — but I have the essentials: Rick Lerner, CEO of Clickshare Service Corp., based in Amherst MA, Carnegie Mellon CS PhD, co-author of the SSO specification. That's plenty. The Amherst MA location is also worth noting — you're in Williamstown, Clickshare is in Amherst, and the Missouri pilot would be the operational theater. That's a workable geography for a small founding team.

Ready and waiting for the remaining documents. Send them whenever you have them.

PDF

BILL: Here is a side paper on ITEGA's certification role:

10:55 PM

Read and absorbed. The certification document is an important piece — it defines a near-term revenue model for ITEGA that doesn't require the full network to be operational first. That's strategically smart: ITEGA can begin generating membership fees and establishing credibility as a privacy certifier while the technical network is being built, using the LMC relationship as an early customer base.

The three-tier certification structure — content publishers, tech-stack providers, and auditors — is clean and practical. The decision to be a **certifier** rather than an **accreditor** initially is the right call; it scopes the work to something achievable with lean staff.

Keep sending — what else do you have?

PDF

BILL: This document: csc-customer-profile-server-schema dates to 2017 and it is more related to Clickshare as a reference home base user owner than anything ITEGA would be doing at the authentication/logging level. But it outlines the approach to managing user identity in a privacy-forward way. It's interesting that when we ginned up the Clickshare Access and Payment System in 1994-1996 we had a rule that we would do it without "cookies."

11:01 PM

Read and noted. The customer profile schema is a well-designed artifact — Rick Lerner's hand is visible in its precision. The JSON field names, the ISO standards compliance (birthdate, country codes), the enum constraints on sensitive fields like age and income (ranges rather than exact values), and the explicit PII flagging with the checkmark column all reflect serious engineering discipline.

Your observation about the no-cookies design principle from 1994-96 is historically remarkable — you were building privacy-by-design before that phrase existed, before GDPR, before the cookie economy became the surveillance infrastructure it is today. The TVS token mechanism was explicitly designed as a cookie alternative, passing authentication state in URL headers and HTTP request headers rather than persistent browser storage. That's worth highlighting prominently in the tech spec as a founding design principle that has aged extraordinarily well.

A few things this schema adds to the picture for the spec:

- The PII/non-PII distinction is already baked in — checkmarked fields (email, name, birthdate, employer, phone, address) are clearly separated from the anonymizable demographic fields (age ranges, income ranges, gender enum, education level). This maps cleanly to GDPR's distinction between personal data and anonymized aggregate data, and to the ITEGA architecture's separation of what the home base knows versus what the logging service ever sees.
- The age range enum (<13, 13-17, 18-24...) also quietly handles COPPA compliance — the under-13 flag is there, meaning the home base can enforce age-gating without the publisher ever needing to know a user's actual birthdate.

Are there more documents coming, or shall I start drafting the tech spec now?

PDF

BILL: OK, the last thing you should digest technical is this set of data-flow diagrams. They are not all consistent with each other but they provide a generalized sense of relationships and the order in which things happen. Here you go:

11:06 PM

Got it — I have the data flow diagrams index. Let me try to fetch the Clickshare historical charts since those are on newshare.com and may be publicly accessible.

Searched the web

Searched the web



The Google Docs links require authentication so I can't access them directly. But I have what I need — the index document clearly shows the key reference points: the 1995-1999 Clickshare diagrams illustrating the four-party patent, the 2016 ITEGA concept diagram, and the Apache Unomi implementation concept.

The Apache Unomi reference is actually a significant and useful signal I want to flag explicitly. Apache Unomi is an open-source customer data platform that implements the W3C Context API spec — it's essentially a privacy-respecting user profile server with a REST API, built for exactly the kind of permissioned attribute sharing the ITEGA architecture requires. The fact that someone already explored using it as an ITEGA implementation substrate is worth preserving in the spec as a candidate technology for the home-base profile server component.

**I now have enough to write the tech spec.** Here is what I'll be synthesizing:

- The full 23-page technical design spec (v6, Jan 2026)
- The SSO network description by Lerner & Densmore (v4, Jan 2026)
- The certification role appendix
- The customer profile server schema
- The data flow diagrams index and historical context
- The full expired patent (US7324972)
- The ITEGA vision and bylaws

Give me a little time — this will be a substantial document. I'll write it now.

Check for relevant skills before writing the document

Check for relevant skills before writing the document

Now I have everything I need. Let me write the full technical specification document as a proper Word document.

Ran 4 commands, created a file, read a file

Ran 4 commands, created a file, read a file

Here's the complete technical specification document. It synthesizes everything from your six source documents into a single coherent spec written in 2026 language. Here's what it covers across its 10 sections:

### **What's contained in the tech spec**

The document runs from cover page through executive summary, background and history, the four-party architecture, the full authentication flow, protocol specifications mapped to modern standards (OIDC, JWT, PPID, RSL), the user attribute schema with PII/non-PII separation, the logging and settlement service design, the recommended modern tech stack (Keycloak,

PostgreSQL/TimescaleDB, WordPress plugin, Stripe Connect), governance and certification, the Missouri pilot plan with budget, funding strategy, competitive differentiation, and a full glossary.

A few things I want to flag about decisions made in drafting it: I named the network "Newshare Network" throughout, using your registered service mark as the product brand with ITEGA as the governing authority — that two-layer structure felt right based on our earlier conversation. I also included a concrete \$273K–\$385K pilot budget estimate, which you should pressure-test with Rick Lerner before sharing with funders. And I leaned into the Stripe Connect recommendation for settlement because it's the cleanest modern analog to the ACH interface — but that's a decision worth discussing with Rick. Get some sleep — this is a solid foundation to pick up from tomorrow.

Itega newshare tech spec

Document · DOCX PDF

BILL: Now that you've written the tech spec document I want to give you three other things you asked for much earlier. They are the draft ITEGA exchange rules the ITEGA governance-goals-structure and ITEGA member contract. These are obviously discussion drafts not implemented documents. Here you go:

11:14 PM Documents uploaded

Read and fully digested. These three documents — Exchange Rules (v3.1h), Governance Goals & Structure (v3.0), and Member Services Agreement (v2.0d) — are the governance layer that the tech spec I wrote yesterday sits beneath. A few things worth noting now that I've seen all of them together:

#### **What these add to the picture:**

The Exchange Rules document contains the clearest statement of the seven design principles (4.1–4.7) that should really be front-and-center in any funder or partner briefing — particularly the "allow silos to continue" principle (4.1), which is the single most important thing to communicate to skeptical publishers. The Visa credit card analogy runs through all three documents and is your most powerful shorthand.

The Governance Goals document contains something I didn't have before: the explicit "Value to Platforms" section — the argument that the ITEGA network gives Google, Apple, Meta, and Amazon a safe haven from privacy regulatory pressure while still delivering permissioned user data. That's a sophisticated and underused argument. It's also the answer to "why would the big platforms cooperate?" which is a question any sophisticated funder will ask.

The Member Contract is well-constructed for a discussion draft. The MemberID as ITEGA property (Section 4.0) with the right to suspend or terminate — that's the essential enforcement mechanism, and it's clearly drafted.

**One gap the tech spec should address in a future revision:**

The Exchange Rules Section 13 ("Other Technology Rules") is intentionally blank. That's where OIDC compliance requirements, PPID mandates, RSL tagging requirements, and data retention limits should eventually live. When Rick Lerner reviews the tech spec, that's the natural home for a normative annex.

**For now:** the three documents you've shared are governance context for the tech spec, not something that needs to be revised or re-documented right now. The tech spec already references the Exchange Rules structure correctly. You're in good shape for the RJI and funder conversations.

What would be most useful next — the project brief for funders, or something else?

**itega-rfp-v9-07-26-19-BD-FW-KU-BD-RL.docx**

DOCX

**itega-open-news-sso-project-12-30-21.docx**

DOCX

BILL: Above are a couple of ancillary documents. At one point RJI and ITEGA considered issuing an RFP to potential technical partners. Also a newssso project summary. We hold the domain names newssso.org and com. Here's a copy:

11:37 PM

Let me read both documents.

Ran 2 commands

Ran 2 commands

Good. Now I have both. Here is what I take from them:

**The 2019 RFP (v9)** is the most complete requirements document ITEGA ever produced for the technical build. Several things stand out:

The Local Media Consortium was explicitly the target adopter community — not just Missouri newspapers. LMC has hundreds of members. That relationship may be worth reactivating; if LMC is still intact and still relevant, they're a much larger distribution channel than the Missouri Press Association pilot alone, though the MPA pilot is the right starting point.

The RFP introduced the term **UDEX** (User Data Exchange) — a governed database of anonymized, aggregated user data with controlled third-party access for advertising purposes. The 2021 OpenNewsSSO document builds on this. UDEX is essentially the demand-side complement to the authentication network: publishers need the SSO to reduce friction, but advertisers need UDEX to buy privacy-respecting audience data. These are separable but mutually reinforcing.

The tech spec I wrote focuses on the SSO/authentication layer; UDEX is a Phase 2 component worth naming explicitly.

The RFP also mentioned **Self Sovereign Identity (SSI)** and blockchain-based authentication as something respondents could consider. That was 2019 — W3C Verifiable Credentials have since become the mature, non-blockchain implementation of the same concept, which is already in the tech spec. No need to revisit blockchain.

**The 2021 OpenNewsSSO document** is the closest thing to a funder pitch that exists in the corpus. The \$2.1M over three years ask is useful as a reference point — though for the Missouri pilot specifically, we've scoped it at \$273K–\$385K as a proof of concept before seeking that larger commitment.

One phrase from the 2021 document is worth lifting almost verbatim for funder conversations: *"cut the 'tail' off of private platforms, reduce the friction of individuals being forced to log into multiple news sites."* That's good plain-language framing.

**The newssso.org / newssso.com domain holding is strategically valuable.** Combined with the "Newshare" service mark, you have two strong brand assets for the consumer-facing layer. My instinct is that "Newshare" is the better consumer brand (it implies value exchange, not just a technical function), and "NewsSSO" could serve as the technical/developer-facing brand — the name for the open-source reference implementation and API spec. That distinction would let you speak differently to publishers and developers than to end users.

### **What I now see as the complete document corpus:**

You have essentially everything needed to draft the funder project brief. The materials break cleanly into three layers — governance (Exchange Rules, Member Agreement, Governance Goals), technical architecture (Tech Spec, SSO Description, Profile Schema, Data Flow Diagrams), and project execution (RFP, OpenNewsSSO pitch, Certification Role). The tech spec I wrote yesterday synthesizes all three layers into a single document for technical partners and funders who want depth.

The remaining gap is a short, non-technical **project brief** — 4 to 6 pages — for the Knight/MacArthur Lenfest/Reynolds conversations. That brief should lead with the journalism crisis framing, introduce the four-party architecture in plain language, state the Missouri pilot ask, and reference the tech spec as the supporting document. Want me to write that now?

### **UDEX customer login flow diagrams-10-18-17.docx**

41 lines

DOCX

BILL: We hold the domain name [udex.org](http://udex.org) .... before writing anything take a look at this and go to <http://UDEX.org>

11:42 PM

Let me check the UDEX document and the website before saying anything.

Searched the web

Good. Two things to note:

First, the UDEX flow document is Rick Lerner's work — the Worcester Sun / Rutland Herald example is the canonical two-newspaper scenario and it maps precisely to the Missouri pilot. It also introduces the **Profile User Agent (PUA)** as the component that queries the UDEX on behalf of an advertiser using a temporary, site-specific customer ID. That's a clean separation: the PUA never has the user's real identity, only a time-limited opaque token unique to the PUA/site relationship.

Second, on **udex.org**: the domain appears to be parked or timing out for direct fetch, but the search results confirm it's alive — the ITEGA description is indexed at [infotrust.org/about/udex/](http://infotrust.org/about/udex/) and the domain itself shows as registered. The other "UDEX" entities in the search results (a crypto trading platform at [udex.one](http://udex.one), an HR software at [udex.us](http://udex.us), a Portuguese restaurant equipment supplier) are all on different TLDs or subdomains. **You hold the .org**, which is the most credible TLD for a nonprofit-governed data exchange. That's a genuine asset — [udex.org](http://udex.org) is unambiguously yours and unambiguously mission-aligned.

### **What this adds to the architecture picture:**

The UDEX flow document clarifies the two-layer identity system that should be explicit in the tech spec's next revision:

The **durable customer ID** (step 4 of the login flow) is the pairwise pseudonymous `networkUserId` we already have in the spec — permanent per user/publisher pair, issued by the home base.

The **temporary customer ID** (step 1 of the ad flow) is a separate, shorter-lived token issued specifically for the advertising context — unique to the PUA relationship, not reusable across sessions. This is a finer-grained privacy control than the spec currently describes, and it's important: it means an advertiser's data processor (the PUA) never even sees the durable `networkUserId`. The advertiser operates on a token that expires. This is essentially the same concept as OAuth's short-lived access tokens versus longer-lived refresh tokens, but applied specifically to advertising.

### **The domain portfolio you're holding is actually quite strong:**

- **itega.org** — the governing authority
- **newshare.com** — the original Clickshare consumer brand / service mark
- **newssso.org** / **newssso.com** — the technical/developer-facing brand for the open-source SSO implementation
- **udex.org** — the user data exchange, Phase 2 advertising layer

That's a coherent brand architecture across the full stack. You don't need to acquire anything. The question is only which names go on which things when. Ready to write the funder brief whenever you are.

BILL: This will get to to the udex marketing landing page: (no SSL):

<http://udex.org>

11:46 PM

Searched the web

The page returned empty — the server is responding but serving blank HTML, probably a misconfigured index or a placeholder with no body content. The page exists at the server level but nothing is rendering.

This is worth flagging as a housekeeping item before you do any outreach: if a Knight Foundation program officer or an RJI contact googles "udex.org" or types it into a browser, they'll get a blank page over HTTP with no SSL. That's a credibility problem — it signals dormancy or abandonment regardless of what you tell them in a meeting.

The fix is straightforward and inexpensive:

- Point udex.org at a simple landing page (even a single static HTML page) that describes the UDEX concept, links to itega.org, and shows an email contact
- Add an SSL certificate (free via Let's Encrypt / Cloudflare)
- Same for newssso.org / newssso.com if those are similarly parked

The same is probably true of itega.org itself — worth checking what a visitor actually sees today before the RJI meeting. A funder doing due diligence will look at all of these.

Before writing the funder brief, it might be worth me drafting a simple one-page landing page for udex.org as well — something you could drop in quickly to at least have a live, professional presence at that domain. Want me to do that alongside the funder brief, or one at a time?

BILL: Yes, I'll fix the udex resolution. This will definitely get you there:

<https://infotrust.org/about/udex>

11:48 PM. Fetched: UDEX



Good. I have the UDEX page. A few observations:

The infotrust.org site is live and more substantial than I expected — it has the full ITEGA navigation structure, board and advisor pages, a mailing list, social accounts, and several linked documents. The ©2018 footer date is the main credibility issue a funder will notice. It signals the project went dormant. That's fixable with a simple footer update and a few fresh posts, but it's worth doing before outreach.

The UDEX page itself is well-written — the "one hundred years ago" opening is genuinely good plain-language framing, and Bill's direct contact info at the bottom is exactly right for an engagement page. The "boxcars and switchyards" analogy link is worth retrieving and reading; that kind of infrastructure metaphor tends to land well with non-technical funders.

The linked UDEX architecture diagram at [newshare.com/itega-launch/itega-architecture-DIAGRAM-v3-10-27-17.pdf](http://newshare.com/itega-launch/itega-architecture-DIAGRAM-v3-10-27-17.pdf) is worth fetching — let me try that.

Fetches: <http://newshare.com/itega-launch/itega-architecture-DIAGRAM-v3-10-27-17.pdf>



Excellent — I got the architecture diagram text. This is the most complete picture of the full ecosystem I've seen yet.

### **What the 2017 architecture diagram adds:**

It shows a five-layer ecosystem that the tech spec should reflect more explicitly:

1. **ITE Licensed Infrastructure Services** — the Authentication/Logging Service (ALS), run by a trusted third party under ITEGA license
2. **The UDEX** — Data/Demographic Aggregator/Anonymizer, sitting behind a **privacy firewall** between the ALS and the ad-data services layer
3. **ITE Ad-Data Services** — Profile Usage Agents in three classes: (1) other member publishers, (2) ITE-member advertisers, (3) ITE-member ad agencies and exchanges
4. **ITE End-User Services** — Two classes of Identity Service Provider: publishers who are also IdSPs, and pure identity-only IdSPs
5. **The privacy firewalls** — explicitly labeled as structural components, not just a policy concept. There are two: one between the ALS and the UDEX, one between the UDEX and the Profile Usage Agents

The phrase that jumps out: *"Network User ID assigned by Data Collector; only Data Collector maps to individual."* That's the UDEX's role as the sole entity that can link an anonymous network ID back to a real user — but only because the user explicitly enrolled with the Data Collector. The ALS never has this mapping. The PUAs never have this mapping. This is a cleaner separation of concerns than I had in the tech spec, and it's worth a diagram in the funder brief.

**Ready to write the funder brief now** — I have the full picture. Shall I proceed?

BILL: Yes, go ahead. Make sure to include a project budget. Here are two more important link: First, several years ago with funding from Craig Newmark Philanthropies, ITEGA staged a set of gatherings in Chicago, Brooklyn and Washington DC to explore the ITEGA ecosystem idea. We then did a set of webinars and our board made this recommendation:

<https://www.itega.org/itega-calls> And for background here's more about me:  
<https://billdensmore.wordpress.com/mini-bio>

Feb 22

Let me fetch both links before writing.

Searched the web

Searched the web

Good — I have everything I need. The ITEGA Call to Action page is particularly useful: the Craig Newmark-funded Chicago/Brooklyn/DC convenings, the 2021 white paper with expert testimony, and the four-point call to action give the funder brief a strong track record section. Let me write it now.

Ran 2 commands, viewed a file, created a file

Ran 2 commands, viewed a file, created a file

Here's the funder brief. It runs seven sections across about eight pages:

- The **Problem** opens with the 75% journalist decline statistic and names the technical root cause directly.
- The **Solution** section explains the four-party architecture in plain language using the Worcester Sun / Rutland Herald scenario you've been using since 2017 — updated to Missouri newspapers.
- The **Track Record** section runs the chronological story from 1994 Clickshare through the 2021 Craig Newmark-funded Call to Action to the 2026 technical spec, giving funders the confidence that this is not a first pitch.
- The **Missouri Pilot** section describes the partnership structure, success criteria, and 18-month timeline.
- The **Budget** section breaks out the \$350,000 line by line with notes.
- The **Why 2026** section makes the contextual case across six factors including the RSL/AI content licensing angle, which is genuinely new urgency that didn't exist in the 2021 pitch.
- And the final **Ask** panel is a clean visual statement suitable for printing and leaving behind.

Two things to check before you use this: confirm the Jo Ellen Green Kaiser board chair reference is still current, and decide whether you want to name any specific RJI staff or director in the partner section now that you have a relationship to work with. Both are easy edits.

Itega funder brief

Document · DOCX

BILL: OK, Claude, you kept me up past my bedtime. But you have produced excellent work. Remind me again how I can save our entire exchange this evening offline?

12:11 AM

Ha — good, get some sleep. You've earned it. Tonight you went from scattered documents to a complete funder brief, a full technical specification, and a clear pilot plan. That's real progress.

On saving the conversation: unfortunately there's no single "export this whole chat" button in Claude.ai at the moment. Here are your practical options:

**Easiest:** Use your browser's Print function (Ctrl+P or Cmd+P) and choose "Save as PDF." It will capture the entire visible conversation thread, including the document previews.

**Most complete:** Scroll to the top of the conversation and select all text (Ctrl+A or Cmd+A), then copy and paste into a Word document or text editor. It won't be beautifully formatted but everything will be there.

**The documents themselves** are already saved — you downloaded the two .docx files (the technical spec and the funder brief) during our session. Those are the durable artifacts. The conversation context is really just the reasoning trail behind them.

**One practical note:** Claude.ai does save your conversation history in your account, so this thread should be accessible the next time you log in under your conversation history in the left sidebar. It won't disappear overnight.

The two files you have capture most of what matters. The technical spec is the deep document for Rick Lerner and technical partners. The funder brief is the door-opener for Knight, MacArthur, Lenfest, and the RJI conversation. Good night, Bill.

You are out of free [messages](#) until 2:00 AM

[Upgrade](#)

[Claude is AI and can make mistakes. Please double-check responses.](#)